Week 7 — Level 2: Intermediate

Files: Saving Your Stuff

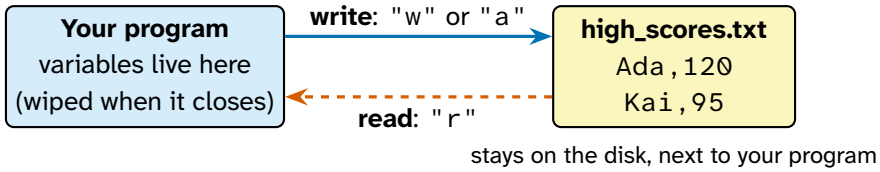
A program forgets everything when it closes, unless it writes things down in a file.

What you will learn

- How to **save** data in a text file, and how to **load** it again next time.
- The three ways to open a file: `"r"` read, `"w"` write, `"a"` append.
- How to clean up lines with `strip()` and cut them into pieces with `split()`.
- What to do when the file is missing, and what a game should never save about you.

The big idea: programs forget, files remember

When a program closes, all its variables are wiped. Your high score? Gone. A **file** is like a notebook in a drawer: it stays on the computer after the program closes, so next time the program can read it again.



Writing to a file

Save one score

```
with open("high_scores.txt", "a") as file:
    file.write("Ada,120\n")
```

- `open(name, mode)` opens the file. `with` closes it again for you when the indented block ends.
- `\n` means “new line”. Without it, everything ends up squashed on one line.

Mode	Means	Like a notebook…
`"r"`	read	Just read it. Crashes if the notebook is not there.
`"a"`	append (add to the end)	Write on the next empty line. Makes a new notebook if needed.
`"w"`	write (replace)	Rip out every page, then write. **Careful: old data is gone!**

Reading a file, line by line

Load the scores back

```
scores = []
with open("high_scores.txt", "r") as file:
    for line in file:
        line = line.strip()            # "Ada,120\n" -> "Ada,120"
        if line != "":                 # skip empty lines
            parts = line.split(",")    # "Ada,120"   -> ["Ada", "120"]
            scores.append([parts[0], int(parts[1])])
print(scores)                          # [['Ada', 120], ['Kai', 95]]
```

Everything you read from a file is **text** (a `str`), just like `input()`. Use `int()` to turn `"120"` into the number `120`.

Bug hunt: the file is not there!

The very first time a game runs, there is no score file yet. Reading it gives:

Output

```
FileNotFoundError: [Errno 2] No such file or directory: 'high_scores.txt'
```

Check the spelling of the name first. If the file really is new, catch the error with `try` and `except`:

```
try:
    with open("high_scores.txt", "r") as file:
        print(file.read())
except FileNotFoundError:
    print("No scores yet - you will be the first!")
```

`try` means “have a go”. `except FileNotFoundError` means “if the file is missing, do this instead of crashing”.

Exercise 7.1: Save High Scores

Folder: `exercises/week07/ex-save-high-scores`. Open `starter.py`, press **Run** twice. Does it remember your first game?

1. TODO 1: in `save_score()`, open the file with `"a"` and write one line `nickname,score`.
2. TODO 2: in `load_scores()`, strip, skip empty lines, split at the comma and append `[nickname, score]`.
3. TODO 3: if the file is missing, make an empty one with `"w"`.
4. Play three times. Open `high_scores.txt` (Finder: double-click it). Copy the first line here:

   _____

5. Check your work with `check_my_work.py`.

Exercise 7.2: Word Guess (choose this one or 7.3)

Folder: `exercises/week07/ex-word-guess`. The secret words live in `words.txt`, one per line.

1. TODO 1: `load_words()` reads every line, strips it, makes it lowercase with `lower()` and skips empty lines.
2. TODO 2: `show_progress()` shows guessed letters and _ for the rest, like `p _ _ e _`.
3. Play a round with your partner, then add three new words to `words.txt`. My new words:

   _____

Exercise 7.3: Dot Art from a File (choose this one or 7.2)

Folder: `exercises/week07/ex-dot-art-file`. Each line of `my_art.txt` is one dot: `x,y,size,color`.

1. TODO 1: `parse_line()` turns `"-150,-60,40,#009E73"` into `[-150, -60, 40, "#009E73"]`.
2. TODO 2: `load_dots()` reads every line that is not empty. Run it: a caterpillar under a sun!
3. TODO 3: `save_dots()` writes every dot back, one per line. Which mode is right here, and why?

   _____

4. Change `my_art.txt` to draw something new. Colors: #E69F00 #56B4E9 #009E73 #F0E442 #0072B2 #D55E00 #CC79A7.

Tip: What should a game save?

A file stays on the computer, and anyone who uses the computer can open it. Save only what the program needs: nicknames, scores, words, drawings. Never save real names, addresses, schools, birthdays or passwords.

Vocabulary box

file Data saved on the computer's disk. It is still there after the program closes.

open / with `open` gets a file ready to use; `with` closes it again automatically.

read / write / append The modes `"r"`, `"w"` and `"a"`: look, replace everything, or add to the end.

strip() Removes spaces and the invisible new-line `\n` from both ends of a string.

split(",") Cuts a string into a list at every comma: `"a,b"` becomes `["a", "b"]`.

FileNotFoundError The error you get when you try to read a file that does not exist.

data privacy Deciding what information about people is saved, who can see it, and keeping it safe.

Reflection

What is the difference between a variable and a file?

My partner's game saved "only the newest score". What do you think the bug was?

Name two things a game should never save about you.

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Files: Saving Your Stuff

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Game Idea Notebook (about 10 minutes)

Folder: `exercises/week07/hw-idea-notebook`. This program keeps a list of your game and art ideas in a file called `game_ideas.txt`. The menu already works; the file part does not.

1. Open `starter.py` and press **Run**. Type `a`, an idea, then `l`. Is the idea listed? (Not yet!)
2. TODO 1: in `add_idea()`, open the file with `"a"` and write the idea plus `"\n"`.
3. TODO 2: in `load_ideas()`, read every line, `strip()` it and keep the ones that are not empty. Catch `FileNotFoundError` and make an empty file, just like in class.
4. Add three ideas, quit with `q`, run the program again and type `l`. They are still there!
5. Open `check_my_work.py` and press **Run**.

Write your favorite idea here. It could become your final project!

_____

★★ Challenge (want more?): Mode detective

Predict what is in `test.txt` after each program runs (start with no file each time). Then try it in Thonny.

Program A

```
for word in ["red", "blue"]:
    with open("test.txt", "w") as file:
        file.write(word + "\n")
```

Program B

```
for word in ["red", "blue"]:
    with open("test.txt", "a") as file:
        file.write(word + "\n")
```

Program A gives: _____ Program B gives: _____

Then add `d` = delete all ideas to your notebook (hint: which mode empties a file?).

Creative (make it yours): Make it yours

Pick one:

- Draw a new picture for `my_art.txt` (Exercise 7.3) on squared paper first: mark each dot's `x`, `y`, size and color. Then type the lines in and run it.
- Make a themed word list for Word Guess: `space.txt`, `animals.txt` or `food.txt`, and change `WORD_FILE` to use it.

Self-check: tick each box when it is true

- ☐ My ideas are still there after I close and run the program again.
- ☐ I can explain the difference between `"w"` and `"a"`.
- ☐ My program does not crash when the file is missing.
- ☐ My files hold ideas, words or pictures only: no real names, addresses or passwords.